

Lecture 08 - Hypothesis Testing

Hypothesis testing is one of the central tools of statistical inference. It is used when we have sample data and want to evaluate a claim about an unknown population parameter or process.

In previous lectures, the focus was on describing data, understanding probability models, studying sampling distributions, and constructing confidence intervals. Hypothesis testing uses the same ideas. The difference is that the reasoning is organized around a decision. We begin with a benchmark claim, measure how far the sample evidence is from that benchmark, and decide whether the evidence is strong enough to reject the benchmark.

This lecture develops the logic of hypothesis testing and then applies it to a worked example about customer waiting times.

1. From Sample Data to Population Claims

In practice, we usually observe a sample rather than a full population. A portfolio manager observes historical returns, not the entire return-generating process. A company observes a sample of customers, not all possible customers. A call center observes a sample of calls, not every call it will ever receive.

The statistical question is therefore:

What can the observed sample tell us about the population or process that generated it?

Confidence intervals and hypothesis tests answer this question in different but related ways.

A confidence interval gives a range of plausible values for an unknown parameter. For example, an analyst may estimate that the expected daily return of a stock lies in an interval around the sample mean.

A hypothesis test focuses on a benchmark value or benchmark claim. For example, an analyst may test whether the expected daily return is greater than zero, or whether average waiting time is above a service target.

The sample is informative, but it is not the whole population. Any conclusion must account for sampling variation.

2. From Confidence Intervals to Tests

Confidence intervals and hypothesis tests are closely connected.

Suppose we construct a 95% confidence interval for a population mean. If a benchmark value is outside that interval, then the benchmark is difficult to reconcile with the sample at the corresponding level. In a simple two-sided test at the 5% significance level, this leads to rejection of the benchmark. If the benchmark is inside the interval, the benchmark is not rejected.

This connection is useful because it shows that a test is not separate from uncertainty measurement. Both tools ask whether the sample evidence is compatible with a parameter value. The interval gives a range. The test gives a decision about a specific benchmark.

3. Null and Alternative Hypotheses

A hypothesis test begins with two statements about the population.

The null hypothesis, denoted H_0 , is the benchmark claim. It is the claim tested against the sample evidence.

The alternative hypothesis, denoted H_1 , is the competing claim. It describes the direction or type of departure that matters in the problem.

For example, suppose a company claims that its average customer waiting time is at most 90 seconds. The benchmark claim is:

$$H_0 : \mu \leq 90.$$

If the concern is that the real average waiting time is too high, the alternative is:

$$H_1 : \mu > 90.$$

Here μ represents the true average waiting time. The null hypothesis represents the company's claim. The alternative represents the concern being investigated.

The test evaluates the sample under the assumption that the null hypothesis is true. If the sample would be unusually far from what the null hypothesis predicts, we reject H_0 . If the sample is reasonably compatible with the benchmark, we fail to reject H_0 .

Failing to reject H_0 does not prove that H_0 is true. It means the sample did not provide strong enough evidence against it under the test being used.

4. One-Sided and Two-Sided Tests

The alternative hypothesis determines the direction of the test.

A one-sided test is used when only one direction is relevant. For example, if a new marketing campaign is useful only if its click-through rate is higher than the old campaign's rate, the alternative is directional:

$$H_1 : p > p_0.$$

Similarly, if a company wants delivery time to be below a target, the alternative may be:

$$H_1 : \mu < \mu_0.$$

A two-sided test is used when departures in either direction matter. For example, suppose a manufacturing process should produce rings with an average diameter of exactly 5 centimeters. A process that produces rings that are too large is a problem, and a process that produces rings that are too small is also a problem. The hypotheses are:

$$H_0 : \mu = 5,$$

$$H_1 : \mu \neq 5.$$

The direction of the test should be determined by the problem before looking at the result. Choosing the direction after seeing the sample would distort the interpretation of the test.

5. Significance Level and Errors

The significance level is denoted by α . It is chosen before running the test.

The significance level controls the probability of a Type I error under the testing procedure. A Type I error occurs when we reject H_0 even though H_0 is true. In practical language, it is a false alarm.

A Type II error occurs when we fail to reject H_0 even though the alternative is true. In practical language, it is a missed detection.

For example, in the call-center case:

1. A Type I error would mean rejecting the claim that average waiting time is at most 90 seconds when the claim is actually true.
2. A Type II error would mean failing to detect that average waiting time is above 90 seconds when it really is above 90 seconds.

Common significance levels are 10%, 5%, and 1%. A smaller α makes rejection harder because the sample result must be more extreme. A larger α makes rejection easier but increases the risk of a Type I error.

The choice of α should reflect the context. A strict threshold may be appropriate when false alarms are costly. A less strict threshold may be used in exploratory settings, but it should still be stated clearly.

6. The Testing Procedure

A simple testing procedure has five steps.

First, state the null and alternative hypotheses. This includes deciding whether the test is one-sided or two-sided.

Second, choose the significance level α before seeing the final test result.

Third, check whether the assumptions behind the test are reasonable. In the simple mean tests used in this lecture, we rely on randomness, independence, and either approximate normality or a sufficiently large sample size.

Fourth, compute the test statistic or the p-value.

Fifth, compare the result with the decision rule and write the conclusion in context.

This structure is important because it separates the statistical calculation from the final interpretation. The calculation tells us whether the sample evidence is unusually strong under the null model. The conclusion connects that decision to the original question.

7. Evidence Relative to Noise

A raw difference between an estimate and a benchmark is not enough.

Suppose the benchmark mean is 90 and the sample mean is 93. The sample mean is 3 units higher than the benchmark. Whether this is strong evidence depends on the standard error.

If the standard error is 1, then the sample mean is 3 standard errors above the benchmark. If the standard error is 5, then the sample mean is only 0.6 standard errors above the benchmark.

The test statistic standardizes the difference:

$$\text{test statistic} = \frac{\text{estimate} - \text{benchmark}}{\text{standard error}}.$$

For a sample mean with known population standard deviation σ , the standard error is:

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The z-statistic is:

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

The z-statistic measures the distance between the sample mean and the benchmark in standard-error units.

A note on proportions. The same testing logic applies to parameters other than a mean. For a population proportion, the benchmark is a proportion p_0 , the standard error is $\sqrt{p_0(1-p_0)/n}$, and the test statistic is again the standardized distance between the estimate and the benchmark. It is worth knowing that this extension exists, because click-through rates, default rates, and pass rates are all proportions. In this course, however, the exercises and the exam focus on tests for a mean. The reasoning developed here transfers directly to proportions when you need it.

8. Critical Values and Rejection Regions

One way to make a decision is to compare the test statistic with a critical value.

The critical value is a cutoff from a reference distribution. In this lecture, the reference distribution is usually the standard normal distribution.

Right-tail test

In a right-tail test, the alternative points to larger values:

$$H_1 : \mu > \mu_0.$$

The rejection region is in the right tail. If $\alpha = 5\%$, the critical value is approximately 1.645. If the observed z-statistic is greater than 1.645, we reject H_0 .

For $\alpha = 1\%$, the right-tail critical value is approximately 2.326.

Left-tail test

In a left-tail test, the alternative points to smaller values:

$$H_1 : \mu < \mu_0.$$

The rejection region is in the left tail. If $\alpha = 5\%$, the critical value is approximately -1.645. If the observed z-statistic is below -1.645, we reject H_0 .

Two-sided test

In a two-sided test, both unusually low and unusually high values count as evidence against the null:

$$H_1 : \mu \neq \mu_0.$$

The significance level is split between two tails. If $\alpha = 5\%$, each tail receives 2.5%. The critical values are approximately -1.96 and 1.96. We reject H_0 if the test statistic is below -1.96 or above 1.96.

9. P-Values

A p-value is another way to summarize the evidence.

The p-value is computed under the assumption that the null hypothesis is true. It measures how unusual the observed evidence is, relative to the null model.

A small p-value means the sample result would be unusual if H_0 were true. A large p-value means the sample result is not especially unusual under the benchmark claim.

The p-value is not the probability that H_0 is true. It is a probability about the data, calculated under the null hypothesis.

The decision rule is:

1. If the p-value is less than or equal to α , reject H_0 .
2. If the p-value is greater than α , fail to reject H_0 .

10. Worked Example: Call Center Waiting Time

Suppose a call center claims that average waiting time is at most 90 seconds. A random sample of 100 calls gives:

$$\bar{x} = 93, \quad s \approx 10, \quad n = 100.$$

We treat the standard deviation as 10 seconds for the purpose of this introductory test.

The hypotheses are:

$$H_0 : \mu \leq 90,$$

$$H_1 : \mu > 90.$$

This is a right-tail test because the concern is that average waiting time is above the claimed limit.

Suppose we use $\alpha = 1\%$. The right-tail critical value is:

$$z_{0.99} \approx 2.326.$$

The standard error is:

$$SE = \frac{10}{\sqrt{100}} = 1.$$

The z-statistic is:

$$z = \frac{93 - 90}{1} = 3.$$

Since 3 is greater than 2.326, the statistic falls in the rejection region. We reject H_0 .

In context, the sample provides strong evidence that the average waiting time is above 90 seconds. This does not prove the exact true average waiting time. It says that the sample mean of 93 seconds is unusually high under the benchmark claim.

11. Statistical Significance and Practical Importance

A statistically significant result is a statement about evidence relative to sampling variability. It is not automatically a statement about business importance.

For example, a very small increase in average return may be statistically significant in a huge dataset. After transaction costs, risk, taxes, or implementation frictions, it may still be economically unimportant.

Conversely, a non-significant result does not prove that there is no effect. The sample may be too small, the data may be noisy, or the test may have low power.

This distinction is essential in finance and business. Statistical significance should be combined with effect size, costs, risk, and practical decision criteria.

12. Writing a Statistical Conclusion

A good conclusion contains three elements.

First, state the statistical decision: reject H_0 or fail to reject H_0 .

Second, translate the decision into the original context.

Third, avoid overclaiming.

For example:

At the 1% significance level, the sample provides strong evidence against the claim that average waiting time is at most 90 seconds. The evidence suggests that the true average waiting time is above the claimed limit.

A weaker or incorrect conclusion would be:

The test proves that the call center is lying.

The statistical conclusion is about evidence under assumptions. It should be direct, but not stronger than the method supports.

13. Summary

Hypothesis testing begins with a benchmark claim and evaluates sample evidence against it. The null hypothesis is the benchmark. The alternative hypothesis describes the departure that matters.

The test statistic compares the sample estimate with the benchmark relative to the standard error. Critical values and p-values provide decision rules. The significance level controls the Type I error probability of the procedure.

The final step is interpretation. We reject or fail to reject H_0 , but we do not prove hypotheses. We use the test to write a careful conclusion about whether the sample evidence is strong enough to challenge a benchmark claim.